Lecture Notes in Production Engineering

Mihai Dragomir · Daniela Popescu · Chin-Yin Huang · Shun-Fung Chiu · Luis Quezada *Editors*

Building Resilience into Production: Contemporary Challenges for the Future

Proceedings of the 27th International Conference on Production Research

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ISSN 2194-0525 ISSN 2194-0533 (electronic)
Lecture Notes in Production Engineering
ISBN 978-3-031-92081-3 ISBN 978-3-031-92082-0 (eBook)
https://doi.org/10.1007/978-3-031-92082-0

This Springer imprint is published by the registered company Springer Nature Switzerland AG
The registered company address is: Gewerbestrasse 11, 6330 Cham, Switzerland

Foreword

The 27th edition of the International Conference on Production Research (ICPR) was held between July 23rd, 2023, and July 28th, 2023, in Cluj-Napoca, Romania, organized by the International Foundation for Production Researched (IFPR) and hosted by the Technical University of Cluj-Napoca (TUCN). The scientific event included captivating keynotes, paper presentations of authors from over 30 countries, a Doctoral Training Workshop, an Early Career Researchers Workshop, the IFPR Women Forum and an Industry Leaders Panel. The overall theme of the conference was "Building resilience into production", and the contributions presented covered a wide range of topics from smart manufacturing to sustainable production.

The current book, included in the Springer series "Lecture Notes in Production Engineering", presents a selection of papers from the ICPR conference, that we hope can be of interest to researchers and practitioners in this field working towards a more resilient and competitive manufacturing sector across the globe. Although regional conditions might differ, the challenges of creating more robust companies and processes, and more innovative and competitive products, while safeguarding the environment and society for the future, bring about exciting and demanding opportunities. Researchers in the field of production engineering and management are called upon to deliver improvements that can contribute to the long-term development of the production sector that has important strategic, economic and social implications.

We express our gratitude as editors, along with the organizers of the conference, to all authors, participants and supporters of the 27th ICPR conference, and we hope that the current volume will represent an important contribution to the knowledge base that IFPR is disseminating internationally for fostering production resilience.

The Editors

Organization

The **27th edition of the International Conference on Production Research** was organized in 2023 in Cluj-Napoca, Romania, by the International Foundation for Production Research and was hosted by the Technical University of Cluj-Napoca, under the theme "Building resilience into production".

Editors

Mihai Dragomir, Daniela Popescu, Chin-Yin Huang, Shun-Fung Chiu, Luis Quezada

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Contents

[21]. Using this method, the melt pool area in pixels is separated from the background by dividing the image into two clusters, Fig. 4.

The dynamical conditions in the melt pool are different at the beginning of the process than at a later stage, which can also be seen in Fig. 4.

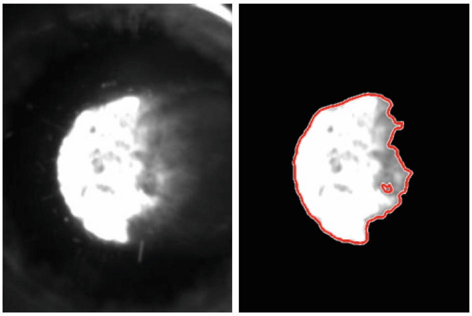

Fig. 4. Image segmentation using the k-means clustering algorithm a) the raw unfiltered image of the melt pool, b) the segmented image of the pool

Also, as mentioned earlier, the laser power is adjusted in the first layers of the sample, which affects the size of the melt pool in this transition phase. After the first 10 layers in structure buildup, the melt pool becomes larger as the process progresses, as represented in Fig. 5. In this paper, the perspective of combining different image segmentation methods to monitor the melt pool geometry is presented. The two methods, which extract two different features of the molten pool, provide valuable information on the process dynamics.

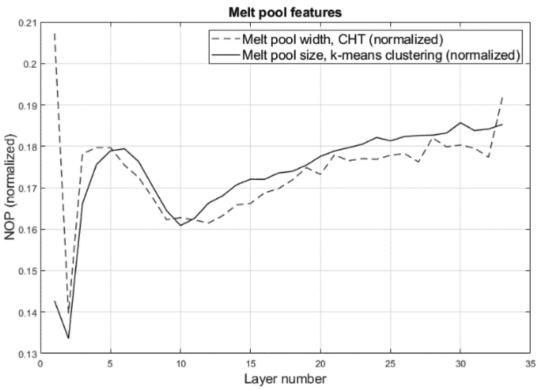

Fig. 5. Melt pool features extracted using CHT and K-means clustering techniques for image segmentation

4 DED-LB Process Outside of the Process Window

In additive manufacturing processes, the production of defect-free parts depends on the used process parameters. Furthermore, the scanning strategy used, especially the degree of overlap between the individual melt tracks, has a significant influence on the density and the geometrical accuracy of the created components.

Another important parameter is the working distance, respectively the focus distance. The working distance is the distance between the nozzle tip and the substrate surface. The focal distance represents the length between the nozzle and the focal plane, which is located just below the substrate surface so that part of the substrate is melted and becomes part of the melt pool. The process parameters influence one another and have a small operating window in which the production of defect-free and dimensionally accurate structures is possible.

In this article, the effect of performing the DED-LB process outside the preset process window was examined, for the use of an insufficient laser power and path distance factor.

Besides an incorrect parameter set, insufficient laser power or heat input during the DED-LB process can result in the collapse of the previously deposited structure. The collapse of the thin-wall structure results from a too-high heat input. The melt pool is deeper and longer, which collapses the structure. After the collapse of the structure the working distance between the nozzle and the substrate surface increases, and the focal plane is raised above the substrate surface. The focal point on the substrate surface thus becomes larger and the heat input decreases and the melt pool is smaller. Less powder will be molten at these positions of the structure. Therefore, the layer thickness is decreasing, and the collapse of the structure is increasing even more until there is no real structure built up.

For the generation of the test samples in the chapters before, a laser power of around 1800–2500 W is used to produce the thin wall structures. These parameters are recommended by the machine manufacturer to produce parts with maximum density and minimum defects. To show the influence of insufficient laser power, a laser power of 500–900 W was used, and cubic structures with an edge length of 25 mm were produced. For this investigation, a path distance factor of 0.6 x width of the deposited melt track was used.

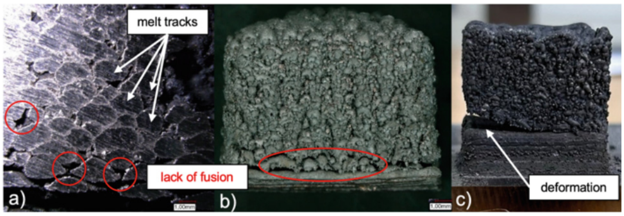

Fig. 6. Lack of fusion in (a) cross-section, (b) surface, (c) connection

The use of insufficient laser power results in the creation of interlayer porosity. The heat input into the substrate and the blow in powder material is too low to create a bonding between the individual melt tracks among each other, which can be seen in

Fig. 6 a) and b) and between the substrate and the deposited structure, shown in Fig. 6 c).

Because of the lack of fusion between the substrate and the deposited structure, the bottom of the cubic parts deforms and raise from the substrate. This deformation results from the thermal stresses generated in the deposited structure due to the high cooling rates during the solidification. When using optimal laser powers, these deformations do not occur because the bottom layers of the structure are firmly fused to the substrate, Fig. 6 c). The lack of fusion is amplified by an increasing path distance factor. The melt tracks don't overlap and lie next to each other. Because of the low laser power and the lack of fusion, unmolten powder particles remain in the structure as pores. A too-low laser power leads to a reduced density. At a path distance of 2 x the width of the deposited track, there is no connection between the tracks. The structure is held together by the melt tracks of the overlying layer.

With the help of the lack of fusion between the melt tracks and the layers among each other, interlayer porosity can be selectively created in DED-LB fabricated structures with the help of too-low laser power. With the help of these graded porous structures, superior device properties can be achieved by fabricating them with locally optimized dynamic and thermal properties, such as a thermo-symmetric structure.

5 Discussion and Conclusions

Even though impressive research work has already been carried out with respect to the detailed analysis of the various process technologies separately, intensive research is still required to achieve a comprehensive understanding of the complex process interrelationships. The ultimate goal is to enable targeted implementation of the process chains in the single-part production of parts with predefined functional properties. One of the most challenging tasks is the implementation of in-situ monitoring systems that enable controlled regulation of the individual process steps. Special potential lies here in the implementation of innovative metrological solutions in combination with various artificial intelligence methods. In this paper, the authors have given insight into some steps of process chain monitoring and observation, such as powder characterization, melt pool monitoring, and additive-subtractive process investigation by forcing the process outside the process window.

Acknowledgments. The presented research results were elaborated within the research projects in AISA (Artificial Intelligence Software Academy) with support from the Ministry of Science, Research and Arts Baden-Württemberg. Furthermore the authors thank the Ministry of Science, Research and Arts of the Federal State of Baden-Württemberg for the financial support of the projects within the *InnovationsCampus Mobilität der Zukunft*. The authors are sincerely grateful for the funding of these research initiatives.

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Simulation-Based Evaluation of Optimized Reconfigurable Cellular Manufacturing Deploying a Diversified Workforce

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Abstract. This paper compares the performance of reconfigurable cellular manufacturing systems (RCMS) and linear systems when employing a diversified workforce with different work speeds.

Due to demographic and labor market trends manufacturing companies increasingly struggle to find skilled personnel for efficient manufacturing operations usually organized in tact-driven production lines. Concepts for flexible manufacturing can foster the integration of a more diversified workforce. Yet, this potential is largely untapped in practice, as the benefits as well as the circumstances under which they can be realized are unknown.

To address this gap, this paper proposes a simulation-based approach to compare the performance of reconfigurable cellular manufacturing systems (RCMS) and linear systems under different scenarios of workforce diversity. The approach is demonstrated within a dynamic multi-variant engine assembly use case, using discrete event simulation. The results confirm the feasibility of the method and point to the advantages of RCMS with different work speeds.

Keywords: flexible manufacturing · planning and control · discrete event simulation · human-centric production

1 Introduction and Motivation

Manufacturing companies face considerable economic and technological challenges, such as volatile markets, unstable economics, increasing competition as well as growing product and process complexity [1]. Societal challenges also include increasing personnel risks due to demographic developments and growing competition in the labor market, as well as changing expectations and fluctuations in the available workforce [2]. Hence, the ability to integrate a diversified workforce with different skills, speeds, strengths and weaknesses is a considerable advantage for manufacturing companies in the future.

M. Dragomir et al. (Eds.): ICPR 2023, LNPE, pp. 336–344, 2025.
https://doi.org/10.1007/978-3-031-92082-0_37

Conventional tact and linear manufacturing systems struggle in this aspect, as different work speeds or worker performance can result in blockages and waiting times and thus lower system Productivity [3]. Flexible manufacturing systems seem to be suited for this, however, specific investigations that quantify such advantage are lacking [4]. Therefore, this paper examines (1) the extent to which workforce diversity affects the productivity of manufacturing systems and (2) the potential advantages of flexible manufacturing systems over traditional production lines in this regard.

The rest of this paper is structured as follows: related work on flexible manufacturing systems, diversity aspects and other relevant workforce-related matters are reviewed in Sect. 2. This is followed by a presentation of the developed simulation approach in the method Sect. 3, followed by a simulation-based evaluation of diversity scenarios in a use-case in Sect. 4. A discussion of the results and an outlook conclude this paper.

2 Related Work

This section highlights related work for two relevant aspects of the concepts behind the scenarios analyzed herein: (1) The characteristics and differences of linear and flexible manufacturing systems and (2) how workforce planning accounts for diversified workforce capabilities, as both are major modeling background for the method and evaluation used in Sects. 3 and 4.

2.1 From Line to Matrix Production

The evolution of manufacturing and assembly systems is driven by partly contradicting requirements such as efficiency and cost pressure, flexibility and changeability as well as fluctuating production volume [5]. Therefore, various manufacturing concepts have been developed, to provide more flexibility, compared to dedicated manufacturing systems, inter alia, single or mixed-multimodel assembly flow lines (AFL). The evolution of flexible systems contains cellular manufacturing systems (CMS), reconfigurable manufacturing systems (RMS) as well as reconfigurable cellular manufacturing systems (RCMS). RCMS is also known as matrix-structured manufacturing systems (MMS) or simply matrix production [6]. A RCMS is characterized by flexibly linked reconfigurable modules, which result in a highly dynamic system that controls and conditionally balances temporary fluctuations in product variant and volume [7, 8]. Its characteristic degrees of freedom of routing and operation flexibilities allow for increased productivity, depending on the flexibility requirements and the actual manufacturing system setup [8, 9]. The concept of RCMS is not only seen as beneficial by decision-makers to tackle current and future challenges [1] but already has been realized in various forms with considerable technological maturity [10].

2.2 Workforce Diversity, Workforce Planning and Workforce Reconfiguration Strategies

In manufacturing, workforce diversity is described by human variability factors, such as age, gender, physical measures, qualification and skills [4]. Factors such as these are important in determining workforce productivity, i.e., the output of goods generated in a given time frame by a specific worker or group of workers [11].

Workforce planning determines on a system level the available workforce capacities as well as the assignment of workers to processes based on human variability factors, particularly skill [12]. Therefore, it includes the problem of process-capacity allocations.

Workforce reconfiguration strategies contemplate the deployment of utility workers, temporary workers, walking workers, bucket brigades or cross-trained workers. They issue optimization measures by addition, removal or interchange of process capabilities and workforce capacities [13, 14].

The aforementioned aspects, i.e. workforce diversity, planning and reconfiguration, have an impact on manufacturing system performance as well as workforce productivity. They can cause significant challenges for resilient system design as well as efficient production planning and control, depending on the labor market situation [2, 14]. Notably, there is little research on how manufacturing systems react to heterogeneous work speeds and workforce productivity by a diversified workforce. This is particularly true for flexible manufacturing systems, such as RCMS, where optimized workforce planning and workforce reconfiguration measures become even more complex, due to individual constraints and additional degrees of freedom [14].

The paper at hand addresses such questions by contributing a simulation-based method to comparatively evaluate the performance of AFL and RCMS concerning workforce diversity. It further demonstrates the synergetic aspects of RCMS with workforce reconfiguration measures (e.g. cross-trained work), to integrate and heuristically optimize a manufacturing system with a diversified workforce. The goal is to improve understanding of the application potential and decision-making on RCMS.

3 Method

This section describes how the linear and flexible manufacturing systems are set up and modeled in a simulation, which will be used in Sect. 4 to compare the system performance when faced with varying work speeds. The comparison is based on a mixed-multimodel engine assembly use case from an industrial partner. The following approach is used to model, compare and evaluate an AFL and an equivalent RCMS in a basic and a diversity scenario. It allows to systematically solve the various decision problems, such as segmentation, allocation or setup, in a sequential manner.

1. Product, process and capacity segmentations
2. Process-capacity and capacity-station allocations
3. Physical setup of stations
4. Heuristic optimization of process-capacity and capacity-station allocations

First, four reference products (V01-V04) are defined to sufficiently cover the existing product spectrum of engines. Methods-time Measurement is used to determine the processing times of all assembly steps. Subsequently, these are aggregated differently to derive the AFL and RCMS with 10 (AFL P01-P10) and 14 (RCMS P01-P14) processes respectively (c.f. Fig. 1). These process segmentations are conducted differently along either organizational (in the case of AFL) or functional, technical and logistical (in the case of RCMS) criteria, as they need to consider matters of tact orientation, line balancing, assembly restrictions, logistical material disposition, technology deployment and tool availability [15]. The aim is to achieve a fair setup for comparison where both settings are designed and optimized accordingly to the respective manufacturing system. The processing times are modeled with stochastic deviations via a triangular distribution of ±10% to represent common human and manufacturing system dynamics. Capacities are modeled for a workforce of ten workers (C01-C10) in both systems. In the use case, the work speed is modeled for every workforce capacity utilizing a linear process-independent factor. Hence, each combination of a process, product and capacity – an action – transforms a specific product based on the individual work speeds of the capacities [2]. The production program consists of a predefined mix of products with a random sequence that resembles actual production sequences in a real-life system. New jobs are released according to the principle of continuous work-in-progress (CONWIP), ensuring an even workload in the system and preventing both overflowing and starving; this is particularly important in the RCMS as routing flexibility can lead to overloading the production system. CONWIP permanently issues eight units of V01 and V02 as well as six units of V03 and V04. Setup processes and capacity or station failure are neglected in the use case, as they would make it harder to identify the effect of the diversity scenario. Evaluating the effects of machine failures on AFL and RCMS has been investigated by the authors in a previous publication [9].

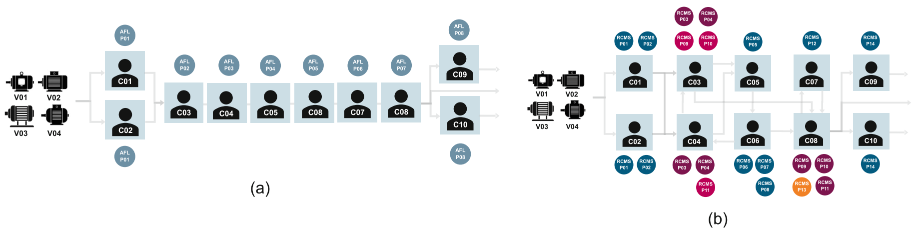

Fig. 1. Setup of the AFL (a) and RCMS (b)

Second, workforce planning assigns initial process-capacity and capacity-station allocations, i.e., how many workers are assigned to certain processes. Like the process segmentations, these allocations are conducted along different criteria. However, there are far more degrees of freedom for the process-capacity allocations in the RCMS, as operation sequences do not constitute a constraint.

Third, the production setup is determined, based on layout restrictions. Both the AFL and the RCMS are given the same space for workstations and work-in-progress.

Fourth, the process-capacity allocations and capacity-station allocations are heuristically optimized in the simulation model. This is an iterative process, with evaluations in the simulation followed by adaptions in the capacity allocations. It is especially important for RCMS where dynamic effects make it less obvious, how and where the operators should be assigned to tasks. Figure 1 outlines the setup of the AFL (a) and RCMS (b).

The AFL, characterized by a sequential arrangement of processes, capacities and stations, only allows for low flexibility (e.g., skipping a workstation if the processes offered by it are not required for a given product type). The RCMS, with a finer process segmentation and refined allocation, can utilize potentials from operation sequence and routing flexibilities. However, due to various entanglements of the assembled product components, none of the products in this case study allow for any operation sequence flexibility; this could potentially increase the flexibility of the RCMS setup if the manufacturing technology enables this. Therefore, in this use case the additional routing flexibility of the RCMS is solely achieved through redundant process-capacity allocations, created by the workforce reconfiguration strategy of cross-training and offering certain processing capabilities by multiple operators and workstations (e.g., Fig. 1, (b) shows how the processing steps RCMS P11 and RCMS P11 are offered by operators C03 and C08, due to cross-training).

The simulation model is implemented in Simo® version 14.22. An easily adaptable RCMS simulation framework was set up for modeling both manufacturing systems in a previous publication by the authors [9]. Additional features, such as a job pool with a queue, which sequences the jobs for release, and linear factors to alter differences in work speed on a process level are added for the experimental study.

4 Evaluation

This section presents the comparative performance analysis for varying work speeds, focusing first on the experimental study with the simulation, followed by an analysis of how the flexibility comes into effect in the RCMS.

4.1 Experimental Study

The modeled AFL and RCMS are validated and evaluated based on reference output, workforce utilization and given constraints with a sample size of ten experiments each. Two scenarios are set up for the AFL and RCMS respectively to analyze the influence of a diversified workforce. The basic scenario features a workforce of ten workers, all at full work speed (100%). The diversity scenario contemplates ten workers, with two workers (C03 and C05) performing considerably slower (70% work speed), while one (C09) is outperforming everyone else (110% work speed). Both the process-capacity and the capacity-station allocations are heuristically optimized for each setup in the simulation model. Table 1 shows the results of the experimental study.

Most notably, the loss of output of the RCMS in the diversity scenario (-5%) equals the total decrease in workforce productivity of the diversified workforce. There are no additional losses in workforce productivity, due to the manufacturing system flexibility. Contrary to that, the major output loss of the AFL in the diversity scenario (-23%) is caused by tact-bleeding, despite heuristic optimization measures on the process-capacity allocations to adjust the system to the altered staff capability. Frequently adjusting a tact-oriented process segmentation or allocation, based on fluctuating workforce diversity levels seems to be unfeasible, due to high planning and implementation expenses. In the use case, the AFL is not able to balance the workload for the diversified workforce and is therefore hampered by productivity losses caused by production system dynamics and (temporary) bottlenecks.

Table 1. Design of experiments and results

	Simulated AFL	Simulated RCMS
Basic Scenario		
Output [Units]	104	105
Workforce Utilization	86%	87%
Diversity Scenario		
Output [Units]	80 (–23%)	100 (–5%)
Workforce Utilization	77% (–9%)	87% (–0%)

Further investigations of the sensitivity on varying work speeds more or less, proved the robustness of the RCMS' performance, while the productivity losses of the AFL remained significant.

4.2 Flexibility Mechanism

The better relative performance of the RCMS in the diversity scenario is founded on refined targeted process-capacity allocations, which affect multiple processes using purposely created process entanglements of first and second grade. Figure 2 outlines these entanglements along capacities and processes as three cross-trained allocations are added. Entanglements of first grade are directly connected to cross-trained allocations, while entanglements of second grade are only indirectly affected. The combination of routing flexibility and such process entanglements between capacities allows for compensation of fluctuating workloads from the production program, mitigates potential efficiency losses of a diversified workforce in a manufacturing environment and overall ensures high capacity utilization.

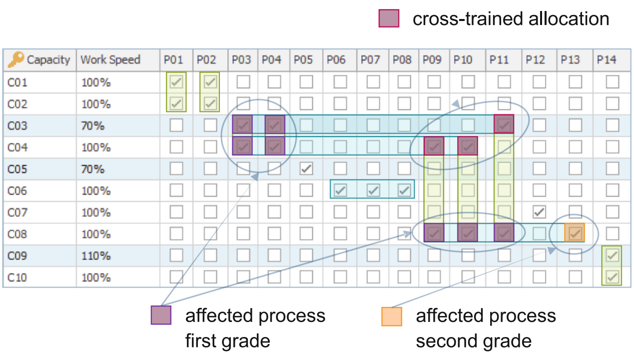

Fig. 2. Process-capacity allocations and entanglements through cross-trained allocations in the RCMS

5 Discussion and Outlook

The results confirm the feasibility of the method. They exemplify how workforce diversity can be modeled and (1) outline the impact of a diversified workforce with varying work speeds on different manufacturing systems' performances in a use case. Furthermore, (2) the presented study showcases that compared to conventional manufacturing, e.g. AFLs, flexible systems, e.g. RCMS', together with combined measures of workforce planning and reconfiguration, e.g. cross-training, allow for optimized integration of a diversified workforce. Consequently, manufacturing system resilience is strengthened and the impact of personnel risks is mitigated, i.e. RCMSs allow for better integration of a wider range of employees without major system productivity loss.

However, there are various limitations to be considered. RMCSs are most likely to outperform conventional AFLs when various constraints regarding market, product and production are existent at the same time. These include short-term market fluctuations in product demand (i.e. volatility and volume) or frequent changes in product design and production process need. Diversity aspects in production, such as persistent work speed differences by a diversified workforce, temporary in-line training measures or the need to integrate performance-diverse technologies or assistance systems, further favor the deployment of a RCMS. For the presented use case, the RCMS most probably does not provide an economic cost advantage, due to increased expenses on material disposition, transport, required personnel coordination efforts as well as hardware-software-integration costs for production planning and system control. Only when high flexibility requirements, as in the diversity scenario apply, the RCMS advantages justify the effort required to make these flexible systems work efficiently in practice. The additional effort includes a digital optimal planning aide (this would in most cases require some form of rolling horizon planning method, e.g. a digital twin with an automatically optimized planning function), cross-training of operators on multiple processes, a flexible intralogistics system that can transport products between stations in a RCMS, some form of visualization and worker-guidance system to help operators navigate their tasks in a system with changing processing requirements. Most of these can be tackled with digitalization, but it reinforces the notion that benefits analysis such as this can only be the starting point of setting up a well-designed RCMS before a practical implementation – this must be followed by detailed planning of the support systems, as described above.

For the AFLs diversity scenario, a workforce reconfiguration strategy, for instance, bucket brigades, can provide benefits on reference output and capacity utilization, however, this particular strategy does have other drawbacks on process responsibility, traceability, tool availability as well as equipment and space utilization. The diversity scenario currently represents a hypothetical scenario, as the use-case partner, as well as most companies, are more likely to use workforce planning measures to recruit workers that can comply with defined productivity levels, rather than to integrate different work speeds. However, the increasingly challenging task of finding and replacing suitably skilled and trainable personnel, also with demographic change, is likely to increase the relevance to integrate a diversified workforce.

This paper uses heuristic rule-based optimization for the aforementioned decision problems on segmentation, allocation and setup. Further technical work can be conducted on advanced planning and control as well as system design, particularly to ensure fast reactivity as well as efficient multi-criteria and energy optimization of RCMS through more sophisticated heuristic condition-based re-routing logics, metaheuristics or reinforcement learning [16].

Aside from the technical aspects, the flexibility considered in this paper also has a pronounced social and human dimension that is not the focus of the current work but should be considered in future research in this field. Two major aspects in this field are considerations of fairness and privacy. The aspect of fairness could bring up questions of payment according to workload vs. payment according to individual ability and its compatibility with the need for economic competitiveness. The privacy concerns include the value of transparent data (and collecting data) on individual skills, abilities and performance to be able to optimally plan with a diverse workforce vs. the need for employees to keep their strengths and weaknesses private to be safe from potential exploitation by the employers. These are just two of the major human factor considerations that should be investigated when looking at implementing measures to include a diversified workforce.

Acknowledgments. The authors would like to acknowledge the financial support of the European Commission for funding the H2020 research project EPIC under grant No. 739592.

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Tailored ALNS to Optimize Real-World Logistic Services for Dependent Patients

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Abstract. In recent years, the number of disabled people have been growing drastically. This trend makes the existing healthcare logistic services inadequate to meet patients demand and, thus, a model to improve the efficiency of this service is necessary. To fulfil this need, the goal of this research is the development of a metaheuristic algorithm to solve a static Dial-a-Ride problem (DARP) with homogeneous requests and a single depot. Indeed, this paper presents the implementation of an Adaptive Large Neighbourhood Search (ALNS) algorithm with some destroys and repair operators tailored for the targeted problem to reach a high-quality solution in an acceptable time consistent with the need of a real healthcare department. This metaheuristic algorithm is compared with the current benchmark in literature and, also, it reports different Key Performance Indicators about the logistic service such as the average time worked and the total waiting time.

Keywords: Dependent patient logistics · healthcare service · adaptive large neighbourhood search · social sustainability

1 Introduction

Today, people with disability represent around 15% of the world population and this number is expected to grow in the next few years due to the increasing of both people aging and noncommunicable diseases [1]. Current healthcare logistic systems are struggling in efficiently manage this growing demand from disabled people. The main reason is that logistic has been always ignored in healthcare despite its relevance and impact in the efficiency of this public service [2]. To improve healthcare systems, such as logistics, mathematical methods have been developed through the years to model specific aspects and simulate different scenarios [3]. Among the healthcare logistic services, patients transportation is one of the most common problems to address which in research is identified as Dial-a-Ride problem (DARP). DARP is a combinatorial problem which differs from the Vehicle Routing Problem since the former optimizes also a social aspect, besides the economic one [4]. To define this social aspect, DARP usually emphasizes the minimization of patients inconvenience, expressed in terms of patients waiting time and ride time, thus neglecting the drivers working conditions which are rarely inserted in the problem objective function [5].

M. Dragomir et al. (Eds.): ICPR 2023, LNPE, pp. 345–353, 2025.
https://doi.org/10.1007/978-3-031-92082-0_38

The goal of this paper is the development of a mathematical model and a tailored metaheuristic algorithm to improve the efficiency of a healthcare logistic service. In particular, this logistic service consists in the dependent patients transportation problem, which well represents a realistic case study based on Austrian Red Cross (ARC) situation. The presented work aims at minimizing the total travel time, but it also analyses the effect of different routing solutions on the patients waiting time and drivers working conditions. The targeted DARP is firstly defined with a mathematical model and, then, solved through a tailored metaheuristic algorithm, whose performance is compared with the literature benchmark.

The paper is organized as it follows. In Sect. 2 a literature review on DARP is presented. Section 3 describes the characteristics of the targeted problem, as well as its objective function while Sect. 4 defines the proposed ALNS algorithm and its operators. Section 5 presents the case study applied to validate and test the developed metaheuristic algorithm and the results obtained. Finally, Sect. 6 draws the conclusions and proposes further research both in the algorithm and in the characteristics of the problem.

2 Literature Review

In literature, the problem of non-autonomous patients transportation falls into the DARP study, defined as the door-to-door transportation of passengers with personalized needs [6]. One of the most common DARP categories is the static version, which does not consider the time in which a request is performed and, thus, it supposes all the requests known in advance [7, 8]. Different types of metaheuristic algorithms have been developed to solve static DARPs. One of the most adopted metaheuristic algorithm is the tabu search, which consists in shaking a reference solution to find a new one, and keep in memory the latest solutions to avoid falling in a local minimum [9, 10]. Other contributions, instead, adopt a deterministic annealing algorithm in which a deteriorating solution can be accepted if the worsening in the objective function is lower than a deterministic threshold dependent on the algorithm advancement [11, 12]. These two types of metaheuristic algorithms are static, i.e. they cannot change during their run according to the past performance. To deal with this issue, novel researches on DARP adopt the Adaptive Large Neighbourhood Search (ALNS) algorithm which includes different destroy-repair operators and, at each iteration, it chooses them with a probability based on their past performance [13]. In [14], ALNS is implemented to solve a DARP and compared with other two metaheuristics to analyse the different performances. ALNS is also applied to solve DARP for a Ride Hailing scenario, i.e. passengers request a vehicle for a single trip from a pickup to a drop-off location, in an American case study [15]. The main drawback of many contributions which adopt ALNS is the implementation of standard destroy-repair operators which does not consider the characteristics of the tackled problem.

To fill these gaps in the literature, the presented work develop and implement an ALNS algorithm with tailored destroy-repair operators to solve a single-depot DARP with homogeneous requests and vehicles which represents a possible real-world case study. In detail, the developed metaheuristic includes some destroy and repair operators which consider the peculiar features of the targeted problem, such as the time windows or the ride time of the patients. In addition, this paper reports some peculiar Key Performance Indicators (KPIs) which measure the service quality for patients and the drivers conditions.

3 Problem Description

The tackled problem regards the dependent-patients transportation from home to an healthcare facility for a medical treatment, or the reverse transportation from the facility to their home. As stated in the first section, this type of problem is called DARP and, in particular, the targeted problem is a static DARP, i.e. all the transportation requests are known in advance. This problem is distinguished by a single depot from and to which each healthcare vehicle starts and ends its journey. In addition, requests are homogeneous since all the patients request the same transportation mode. Consequently, the fleet of vehicles is homogeneous too, all including the same pieces of equipment needed to fulfil the placed requests and all distinguished by the same capacity. Each healthcare vehicle is assigned to a single driver who is distinguished by a fixed working shift which cannot be violated. Each transportation request includes one patient and it is characterized by a time window within which the vehicle should arrive at the assigned location. In detail, these time windows cannot be violated on the right-hand side, i.e. vehicles cannot arrive after the latest arrival time. However, the early arrival is allowed and, in this circumstance, the vehicle would wait at the assigned location until the earliest arrival time. Thus, if the vehicle is carrying some patients, they would also wait outside the destination until the earliest time of its time window is reached.

Regarding the objective function, it aims at minimizing the total travel time spent cs in the solution s of this transportation service. Since the metaheuristic algorithm could provide infeasible solutions during the search, an evaluation function fs is computed to compare the different solutions (Eq. 1). This function includes, besides the objective function, the total violation of load (qs), ride time (rs), shift (ds) and time windows (ts). All these violations are weighted through dynamic parameters (respectively α, β, γ and ε) which are initialized with value 1 and increase or decrease of a value δ at each algorithm iteration. In detail, if solution s violates a specific constraint, the corresponding value is multiplied by δ, otherwise it is divided by it.

$$f\mathrm{s} = c\mathrm{s} + \alpha * q\mathrm{s} + \beta * r\mathrm{s} + \gamma * d\mathrm{s} + \varepsilon * t\mathrm{s} \tag{1}$$

It is important to clarify that a solution can be selected as the final best one only if it is feasible, i.e. $f(s) = c(s)$.

4 Metaheuristic Algorithm

Due to its several features and constraints, the targeted DARP is a NP-hard problem and, thus, it cannot be solved optimally for large instances. For this reason, an ALNS algorithm has been developed and implemented with tailored destroy and repair operators to solve this problem and find an efficient global best solution (*s_best*). ALNS is a metaheuristic algorithm based on different destroy and repair operators which are alternatively chosen at each iteration to provide a new solution. The peculiarity of this method is that the operators selection is adapted to their past performances.

In detail, after the generation of an initial reference solution s^* through a construction heuristic (described in Sect. 4.1), a destroy-repair operators pair is selected to modify s^* and find a new solution s'. If the new solution is promising, i.e. cs' at most 2% worse than cs*, an intra-route local search is applied to further improve its quality. Then, an acceptance criteria (described in Sect. 4.3) is checked to measure the performance of the destroy and repair operators selected. The selection of the these operators is defined by a roulette wheel based on the weight w of each one (Eq. 2).

$$w\mathrm{i} \,/ \sum w\mathrm{k} \tag{2}$$

where i represents the specific operator and k iterates over all the operators. In detail, at each iteration, according to the performance of the new generated solution, the score π of the adopted operators is updated with one of the following values:

- $\sigma 1 \rightarrow$ if the operators pair provides a new global best solution
- $\sigma 2 \rightarrow$ if the operators pair provides a new reference solution, but not best.
- $\sigma 3 \rightarrow$ in all the other circumstances

Every constant time interval *t_seg*, dependent on the dimension of the studied instance, the operators weights are updated through the following equation (Eq. 3):

$$w\mathrm{i,j+1} = w\mathrm{i,j} \,*\, (1-\rho) + \rho * (\pi\mathrm{i}\,/\theta\mathrm{i}) \tag{3}$$

where i is the specific operator, j represents the time intervals in which the algorithm is divided, ρ is the reaction factor and θ is the number of times the specific operator i is selected in the current time interval j.

4.1 Initial Solution

The construction heuristic starts by sorting the transportation requests according to their time windows, putting first the earliest requests. Then, following this order, one request per route is inserted. The remaining ones are sequentially inserted in the route which minimizes one of the following four distances, randomly selected for each request individually:

- Distance between the pickup of the last request and the pickup node of the new one
- Distance between the pickup node of the last request and the delivery node of the new one
- Distance between the delivery node of the last request and the pickup node of the new one
- Distance between the delivery node of the last request and the delivery node of the new one

From the description of this construction heuristic, it is deducted that the generated initial solution can be infeasible with respect to time windows or shift violations.

4.2 Destroy and Repair Operators

The body of the developed ALNS is the implementation of different destroy and repair operators, including some general-purpose and some tailored for this specific DARP.

The destroy operators remove a specific number of requests which is randomly selected between a range defined by the contribution of [16]:

- Random removal: it removes random requests from random routes.
- Worst removal: it removes the worst requests in terms of evaluation function. In particular, it sequentially removes the request that provides the largest decrement in the evaluation function after its removal.
- Related removal: this operator displaces the first request randomly. Then, it removes the requests most related with the already-removed ones, where the relatedness R of two requests is defined by the travel time between them and the similarity of their time windows.
- Load violation removal: this operator removes the requests distinguished by the largest load violation. If it is null, the requests which generates the largest reduction of the objective function are selected for removal.
- Time-based removal: this operator removes the requests which cause the largest shift violation. If there is no shift violation, the requests distinguished by the largest violation of ride-time are displaced. If ride-time violation is null too, the requests which generates the largest reduction of the objective function are selected for removal.

The repair operators insert each request removed by the destroy ones:

- Best insertion: this operator inserts the requests in the order with which they are removed by the destroy operator. It finds the three best positions among all the routes for the current request and it selects one of them through a roulette wheel selection. The positions are evaluated in terms of evaluation function increase and, thus, the best position is the one that increase this function the least.
- Random insertion: same as "Best insertion" but here the requests are selected randomly from the list of removed requests.
- Regret-2 insertion: for each removed request, this operator finds the two best positions and, then, it computes the regret value as the difference in the evaluation function between these two positions. Finally, this operator selects the request which maximize this value.
- Zero-load insertion: it creates a list of positions in which the vehicle load is null. Then, this operator chooses randomly one of these positions and inserts a random request there. After each insertion, the new zero-load positions are updated.

4.3 Acceptance and Stopping Criteria

As described in the first paragraph of Sect. 4, at each iteration the ALNS generates a new solution *s'* which is compared to the reference one s^* and to the best one *s_best*. In detail, if the new solution is feasible and better than the best, i.e. fs' $< f$s_best, the former becomes the new global best solution. In addition, if *s'* is better than the s^*, it also becomes the new reference solution. On the contrary, if no improving solutions are found for a time *t_last*, which depends on the dimension of the considered instance, *s'* becomes the new reference solution in any case to shake the solution and move from the local optimum.

The developed ALNS stops when the run time reaches the time limit *t_tot*, which is based on the number of nodes of the tackled problem, and it provides the global best solution found up there.

5 Case Study and Results

The proposed DARP is a representation of a real-world scenario which can occur in the healthcare transportation service of ARC. However, to validate and test the presented work, the developed metaheuristic algorithm is implemented with the instances presented by [17]. In addition, Table 1 lists the values of the constant problem parameters adopted in the aforementioned case study.

Table 1. ALNS parameters

Parameter	Value
$\sigma 1$	10
$\sigma 2$	2
$\sigma 3$	0
ρ	0.1
δ	UNIF(0.05–0.1)

The results obtained with the developed ALNS are compared to the ones of [17] which represents the current benchmark for this type of DARP. As shown in Table 2, the performance of the proposed metaheuristic algorithm is similar to the benchmark one since it is at most 5% worse in 18 out of 20 tested instances. Indeed, in two of the biggest instances (marked in bold type in Table 2) the developed ALNS is still distinguished by inefficient results which need to be improved.

Besides the comparison with the current benchmark, this paper analyses two performance indicators related to service quality and drivers condition. The first indicator is the average waiting time for each patient which provides a quantitative measure on patients satisfaction (Fig. 1). In this problem, the average waiting time varies drastically among the different instances, moving from almost 1 min/patient to more than 13 min/patient.

The second evaluated aspect is the time worked by drivers to quantitatively measure their working conditions. In detail, for each instance the minimum, average and maximum time worked by drivers are computed to study if the workload are balanced among them (Fig. 2). Results show that, apart from the first instance, the other ones are quite balanced since the range box are not widely spread, bounded between 200 min/day and 480 min/day.

Table 2. Comparison between proposed ALNS and current benchmark in all the instances of [17]

Instance	N° of requests	fs_best benchmark	fs_best ALNS	% worse
R1a	24	190.02	190.02	0.0
R2a	48	301.34	302.49	0.38
R3a	72	533.01	543.31	1.93
R4a	96	573.92	589.43	2.70
R5a	120	636.77	655.43	2.93
R6a	144	802.12	836.14	4.24
R7a	36	291.71	291.79	0.03
R8a	72	490.58	513.70	4.71
R9a	108	666.96	682.80	2.37
R10a	144	866.97	903.53	4.22
R1b	24	164.46	164.46	0.0
R2b	48	296.65	303.44	2.29
R3b	72	490.16	510.56	4.16
R4b	96	533.15	558.31	4.72
R5b	120	583.12	610.76	4.74
R6b	**144**	**742.12**	**796.39**	**7.29**
R7b	36	248.21	248.47	0.10
R8b	72	462.50	470.97	1.83
R9b	108	600.18	627.93	4.62
R10b	**144**	**796.90**	**876.98**	**10.05**

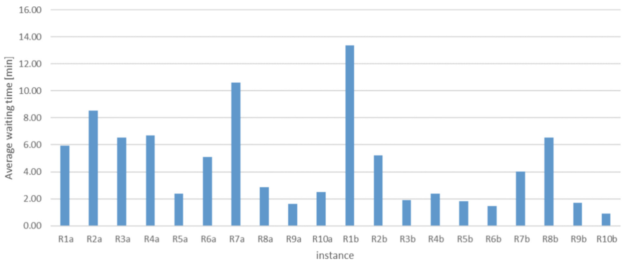

Fig. 1. Average waiting time in each tested instance

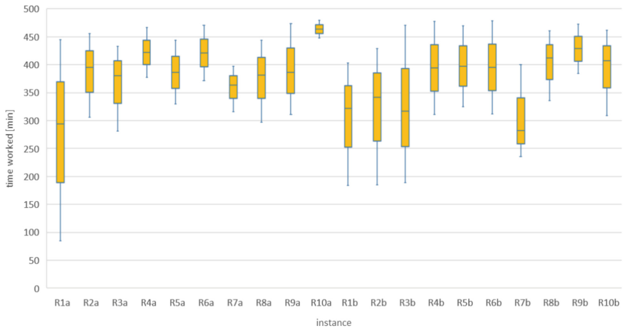

Fig. 2. Range box of the time worked by drivers in each tested instance

6 Conclusions

In the presented paper, a realistic transportation problem of non-autonomous patients is tackled. In detail, this problem is solved as a single-depot DARP with homogeneous requests and vehicles, with the objective of minimizing the total travel time. The peculiarity of this work is, firstly, the consideration of additional aspects in the solution analysis, such as the patients waiting time and the drivers conditions in terms of time worked. Another novel characteristic of this research is the implementation of an ALNS metaheuristic algorithm based on tailored destroy and repair operators, which is adopted to solve large instances of the targeted problem. The developed metaheuristic presents performances similar to the ones of the literature benchmark (at most 5% worse) for 18 out of 20 tested instances. Furthermore, results show that the patients average waiting time ranges between 1 and 13 min/patients among the different instances while the time worked by drivers is quite balanced between the least and the most employed workers.

Further research can be carried out about this type of transportation problem. First of all, the developed ALNS should be improved to be efficient also in the biggest instances of the literature benchmark. Furthermore, additional features can be applied to the targeted problem, such as heterogenous requests or multiple depots, to make it closer to the reality. Finally, the developed ALNS can be tested and validated with a real dataset provided by specific healthcare department, such as the ARC, to analyse its performance in a real-world scenario.

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